

London NRMM LEZ

Compliance Effect Analysis

Data period: 2016–2024

n = 10,749 audit records across three equipment groups

Research Plan v1.18 | 31 March 2026

Executive Summary

Three-component decomposition of stage-based compliance improvement into natural market turnover, voluntary proactive LEZ response, and direct enforcement.

Principal findings

- **59.7%** of all audit records are self-compliant at initial audit — no enforcement needed
- **22.2%** are driven compliant via *registration* only — no stage change results
- Only **8.7%** involve a machine below the stage threshold that is removed or driven compliant — the records that produce actual emissions change
- Stage non-compliance rate: **13.8%** of warm fleet vs 29.6% from `Initial.Machinery.Compliance` — gap is entirely registration-only failures
- **Rest of London Phase B** stage $\bar{e} = 4.0\%$ — the RoL fleet was already largely above IIIB threshold; its 28.1% `Initial.Machinery.Compliance` rate reflects unregistered machines, not emissions failures
- Voluntary proactive LEZ response ($\lambda_{\text{Proactive}}$) is detectable in all groups when estimation is restricted to self-compliant warm records

Data and Fleet Structure

Equipment groups

Group	Description	Phase B threshold
Constant Speed (CS)	Generators and constant-speed plant	Stage V (≥6)
CAZ+	Variable-speed, Central Activity Zone + Olympic Area	Stage IV (≥5)
Rest of London (RoL)	Variable-speed, Greater London outside CAZ+	Stage IIIB (≥4)

Stage mapping: I=1, II=2, IIIA=3, IIIB=4, IV=5, V=6, ZE=7

Model phases

Phase	Period	Key change
A1	1 Jan 2016 – 31 Dec 2018	Initial programme
A2	1 Jan 2019 – 31 Aug 2020	Stage V equipment newly available
B	1 Sep 2020 – 31 Dec 2024	Threshold step-up: CS→V, CAZ+→IV, RoL→IIIB

Phase B subdivided into three equal segments (~527 days) for variable-speed estimation.

Compliance Routing — Classification Rules

Every audit record assigned to one of eight outcomes using three fields:

`Initial.Stage` vs threshold · `Initial.Machinery.Compliance` · `Final.Machinery.Compliance`

Routing	Stage OK?	Initial.Machinery.Compliance	Final outcome	Stage change?
Self-compliant	✓	compliant	—	No
Compliant — exemption/viability	✗	compliant	—	No
Driven compliant — registration	✓	non-compliant	compliant	No
Removed — registration	✓	non-compliant	removed	No
Non-compliant — registration	✓	non-compliant	unresolved	No
Driven compliant — emissions	✗	non-compliant	compliant	Yes
Removed — emissions	✗	non-compliant	removed	Yes
Non-compliant — emissions	✗	non-compliant	unresolved	No

Only the two **bold** rows produce a stage transition through enforcement.

Compliance Routing — Overall Counts (n = 10,749)

Routing	n	%
Self-compliant	6,417	59.7%
Driven compliant — registration	2,388	22.2%
Removed — emissions	557	5.2%
Driven compliant — emissions	380	3.5%
Compliant — exemption/viability	374	3.5%
Non-compliant — registration	356	3.3%
Non-compliant — emissions	216	2.0%
Removed — registration	61	0.6%

Registration-driven routes together: **26.1%** of all records — none produce a stage change.

Emissions-driven enforcement (stage change): **8.7%** (937 records).

Compliance Routing — Year-by-Year

Cold fleet (top) · Warm fleet (middle) · All records (bottom)



Decomposition Framework

$$\lambda_{\text{Policy}} = \lambda_{\text{CF}} + \lambda_{\text{Proactive}}$$

Three components

Component	Symbol	Estimated from
Counterfactual rate	λ_{CF}	Cold-engaged fleet stage distribution
Policy rate	λ_{Policy}	Warm fleet, self-compliant records only
Proactive rate	$\lambda_{\text{Proactive}}$	Derived: $\lambda_{\text{Policy}} - \lambda_{\text{CF}}$ (floored at 0)

Enforcement exposure

$$\bar{e}_{g,\phi} = \frac{\#\{records : \text{Initial.Stage} < \text{threshold}_{g,\phi}\}}{N_{g,\phi}^{all}}$$

Counts records where stage is below threshold across **all records** (cold + warm). Registration-only non-compliance excluded by construction.

WLS estimator

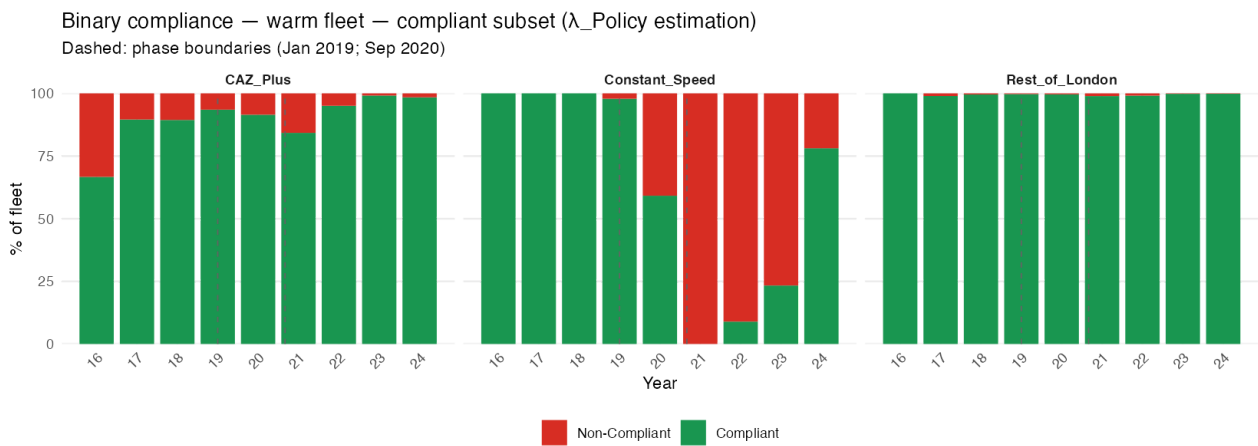
$$\frac{\Delta p_{c,t}}{\Delta t} = \lambda \cdot (1 - p_{c,t})$$

Cold Fleet — Compliance Profiles



Policy Rate (λ_{Policy}) — Self-Compliant Warm Subset

Estimated from warm records that are **self-compliant at initial audit** — machines meeting stage threshold, registered. Captures voluntary proactive LEZ response free from enforcement contamination.



λ_{Policy} estimates

Case	n	λ_{Policy}	SE	95% CI	Significant?
A1/A2 Variable Speed pooled	2,094	0.361	0.162	[0.043, 0.679]	Yes
CS Phase B	252	0.264	0.182	[−0.093, 0.620]	Marginal
CAZ+ Phase B	1,416	0.536	0.024	[0.490, 0.582]	Yes — high precision
RoL Phase B	2,298	0.338	0.177	[−0.010, 0.685]	Marginal

Counterfactual Rate (λ_{CF})

λ_{CF} = annual rate of stage improvement without LEZ intervention.
Estimated from the cold-engaged fleet (sites not registered with the LEZ).

Stream B granular estimates (authoritative)

Group	λ_{CF}	SE	95% CI
Constant Speed	0.358	0.067	[0.226, 0.490]
CAZ+	0.363	0.052	[0.261, 0.464]
Rest of London	0.219	0.048	[0.125, 0.313]

Bias direction

The cold fleet **overestimates** the true counterfactual rate — LEZ demand elevates second-hand stock entering unregistered sites. λ_{CF} is a conservative upper bound; $\lambda_{Proactive}$ is a conservative lower bound.

CS Phase B: $\lambda_{CF} = 0$ confirmed by inspection — no Stage V machinery in cold CS fleet 2021–2023.

Full Decomposition

$$\lambda_{\text{Proactive}} = \lambda_{\text{Policy}} - \lambda_{\text{CF}} \quad (\text{floored at } 0)$$

Case	λ_{CF}	λ_{Policy}	$\lambda_{\text{Proactive}}$	$\bar{e}$	$\lambda_{\text{Pro}} / \lambda_{\text{Policy}}$	Reliability
A1/A2 Var Speed	0.130	0.361	0.231	0.076	64%	Good
CS Phase B	0.000	0.264	0.264	0.711	100%	Moderate
CAZ+ Phase B	0.161	0.536	0.375	0.199	70%	Excellent
RoL Phase B	0.152	0.338	0.186	0.039	55%	Moderate

CS Phase B: $\lambda_{\text{CF}} = 0 \rightarrow$ every unit of voluntary compliance is entirely attributable to the proactive LEZ response.

CAZ+ Phase B: most precisely estimated case; 84% $\rightarrow$ 98.7% compliance in self-compliant warm subset over Phase B; proactive response (0.375/yr) is the largest of any group.

RoL Phase B: genuine but modest proactive response (0.186/yr) against very low stage-threshold enforcement exposure — policy effect here is predominantly through registration compliance, not stage upgrading.

Key Results Summary

Finding	Value
Self-compliant at initial audit (all records)	59.7% (6,417)
Driven compliant — registration (no stage change)	22.2% (2,388)
Driven compliant or removed — emissions (stage change)	8.7% (937)
Persistent non-compliant	5.3% (572)
Stage non-compliance rate — warm fleet	13.8%
λ_{CF} — Constant Speed	0.358/yr [0.226, 0.490]
λ_{CF} — CAZ+	0.363/yr [0.261, 0.464]
λ_{CF} — Rest of London	0.219/yr [0.125, 0.313]
λ_{Policy} — CAZ+ Phase B	0.536/yr [0.490, 0.582]
$\lambda_{Proactive}$ — CAZ+ Phase B	0.375/yr (70% of λ_{Policy})
$\lambda_{Proactive}$ — CS Phase B	0.264/yr (100%; $\lambda_{CF} = 0$)
$\bar{e}$ — CS Phase B (highest)	0.711
$\bar{e}$ — RoL Phase B (lowest)	0.039